

One-Way Functions, Hardness Concentration Bits, and PRGs

CS 171: Cryptography

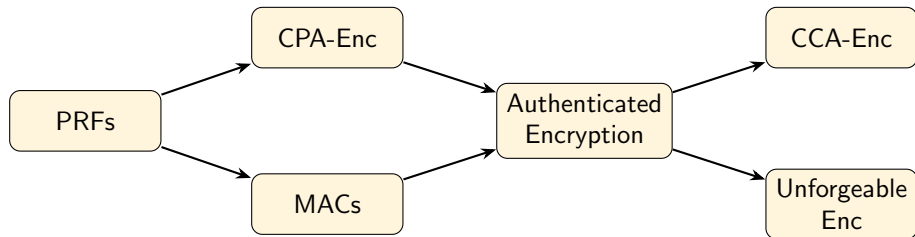
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October 8, 2026

Overview

- 1 One-Way Functions
- 2 Hardness Concentration
- 3 PRG from Hardness Concentration
- 4 Getting Beyond One Bit
- 5 Summary

What We Have Seen So Far



Question

Everything rests on PRFs. **How can we construct PRFs?**

Constructions of PRFs / Block Ciphers

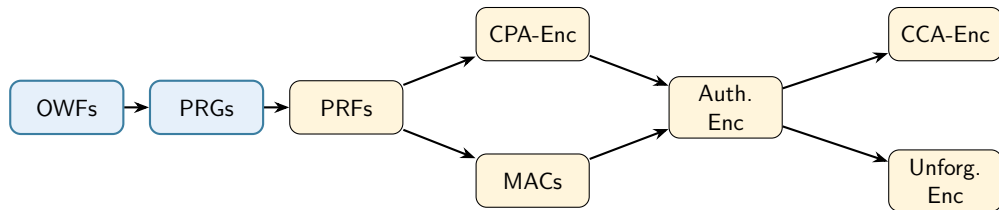
Practical constructions

A bit of an **art**! AES, DES (lecture 8): heavily analyzed, but heuristic.

Theoretical constructions

A **rigorous** approach! Built from simpler, well-defined assumptions, with proofs.

One-Way Functions



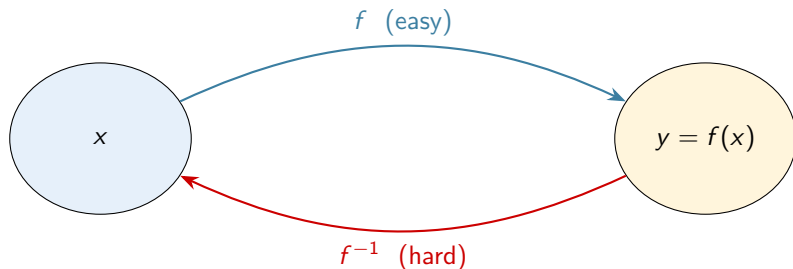
The foundation

Everything we have built can be based on **one-way functions**. More accurately:
 $\text{OWFs} \Rightarrow \text{PRGs} \Rightarrow \text{PRFs} \Rightarrow \dots$

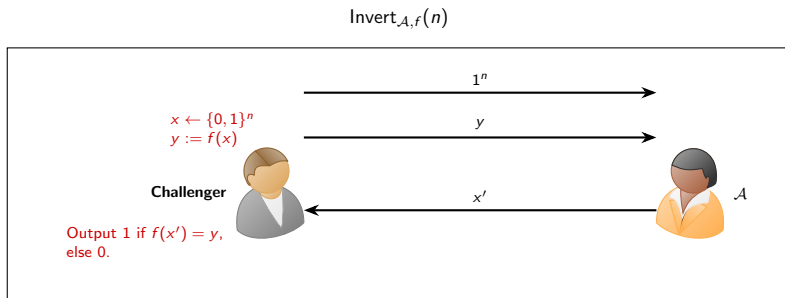
Define: One-Way Functions

Informal definition

A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ that is **easy to compute** but **hard to invert**.



Invert $_{\mathcal{A},f}$: Challenger vs. Adversary



Remarks

- 1 $\mathcal{A}$ need not find x itself: **any** preimage x' of y wins.
- 2 The probability is over the choice of x (and $\mathcal{A}$'s coins).

One-Way Functions: Formally

Game $\text{Invert}_{\mathcal{A},f}(n)$

- 1 $x \leftarrow \{0, 1\}^n$, $y := f(x)$. $\mathcal{A}$ is given 1^n and y .
- 2 $\mathcal{A}$ outputs x' .
- 3 Output 1 if $f(x') = y$, and 0 otherwise.

Definition 1 (One-way function)

A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is a **one-way function** if:

- **(easy to compute)** There is a polynomial-time algorithm M_f with $M_f(x) = f(x)$ for all x .
- **(hard to invert)** For all PPT $\mathcal{A}$: $\Pr [\text{Invert}_{\mathcal{A},f}(n) = 1] \leq \text{negl}(n)$.

Is g a OWF? (Change One Input)

Given: f is a OWF

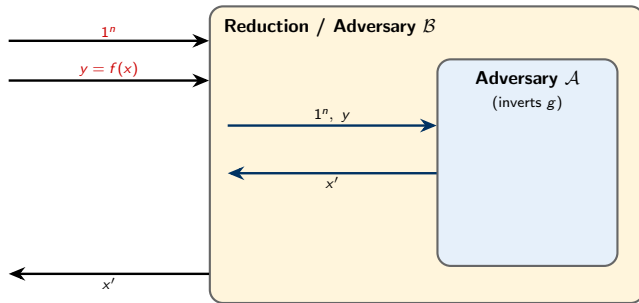
Define

$$g(x) = \begin{cases} f(x) & \text{if } x \neq 0^n, \\ 0^n & \text{if } x = 0^n. \end{cases}$$

Yes!

g is easy to compute, and g differs from f only on $x = 0^n$, which the challenger samples with **negligible probability** 2^{-n} .

g Is One-Way: Proof by Reduction



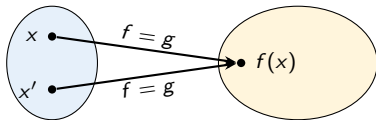
Given / To prove

Given: $\Pr [\text{Invert}_{\mathcal{A},g}(n) = 1] \geq \varepsilon(n)$ for non-negligible ε .

To prove: $\Pr [\text{Invert}_{\mathcal{B},f}(n) = 1] \geq \delta(n)$ for non-negligible δ .

g Is One-Way: Three Cases

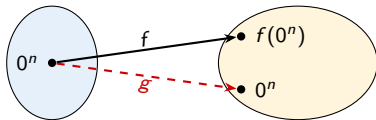
Suppose $\mathcal{A}$ wins: $g(x') = g(x)$.



Case 1: $x \neq 0^n$ and $x' \neq 0^n$

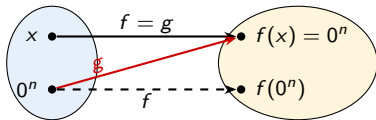
f, g coincide on x and x' :

$f(x') = g(x') = g(x) = f(x)$. So $\mathcal{B}$ **wins**.



Case 2: $x = 0^n$ (probability 2^{-n})

$\mathcal{A}$ is given $f(0^n)$, not $g(0^n) = 0^n$: not its own game, **no guarantee**.



Case 3: $x \neq 0^n$ and $x' = 0^n$

Needs $f(x) = g(0^n) = 0^n$ (probability p). $\mathcal{B}$ outputs 0^n : a g -preimage, but in general **not** an f -preimage.

g Is One-Way: Putting It Together

When can $\mathcal{B}$ fail although $\mathcal{A}$ wins?

Only in case 2 ($x = 0^n$, probability 2^{-n}) or case 3 ($f(x) = 0^n$, probability $p := \Pr_x[f(x) = 0^n]$).

Success probability of $\mathcal{B}$

$$\begin{aligned}\Pr[\text{Invert}_{\mathcal{B},f}(n) = 1] &\geq \Pr[\text{Invert}_{\mathcal{A},g}(n) = 1] - \Pr[x = 0^n] - \Pr[f(x) = 0^n] \\ &= \Pr[\text{Invert}_{\mathcal{A},g}(n) = 1] - 2^{-n} - p \\ &\geq \varepsilon(n) - 2^{-n} - p.\end{aligned}$$

Why p is negligible

The adversary that outputs a uniform x' wins Invert_f with probability $\geq \Pr[f(x) = 0^n \text{ and } f(x') = 0^n] = p^2$. As f is a OWF, p^2 , and hence p , is negligible.

So $\varepsilon(n) - 2^{-n} - p$ is non-negligible, contradicting that f is a OWF.



Is g a OWF? (Drop the Last Bit)

Question

If f is a OWF and $g(x) = f(x)_1 \cdots f(x)_{|f(x)|-1}$ (the output of f with its last bit dropped), is g a OWF?

Two possible answers to argue:

- **Yes:** for every OWF f , g is a OWF.
- **No:** there exists a OWF f such that g is not a OWF.
 - Assume a OWF h and use it to construct f .
 - Prove f is a OWF; prove g is *not* a OWF.

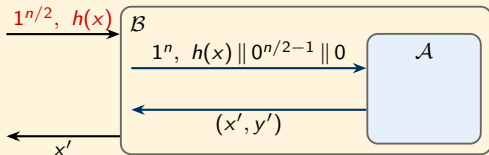
Is g a OWF? (Answer)

No!

Let $h : \{0, 1\}^{n/2} \rightarrow \{0, 1\}^{n/2}$ be a OWF, and define, for $x, y \in \{0, 1\}^{n/2}$,

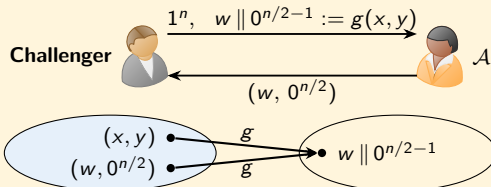
$$f(x, y) = \begin{cases} h(x) \parallel 0^{n/2-1} \parallel 0 & \text{if } y \neq 0^{n/2}, \\ x \parallel 0^{n/2-1} \parallel 1 & \text{if } y = 0^{n/2}. \end{cases}$$

f is one-way: reduction to h



Why: the challenge ends in 0, so a correct answer has $y' \neq 0^{n/2}$ and $h(x') = h(x)$. $\mathcal{A}$ sees its own game unless $y = 0^{n/2}$: $\Pr[\mathcal{B} \text{ wins}] \geq \Pr[\mathcal{A} \text{ wins}] - 2^{-n/2}$.

g is not one-way: the attack



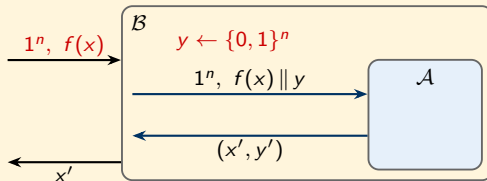
$g(w, 0^{n/2}) = w \parallel 0^{n/2-1}$: a collision, so $\mathcal{A}$ wins with probability 1.

The Challenge: Is g a OWF? (Concatenate Input)

Question

If f is a OWF and $g(x, y) = f(x) \parallel y$, is g a OWF?

Yes! Reduction to f



Why: $\mathcal{A}$'s input is distributed exactly as in its own game. If $\mathcal{A}$ wins, $f(x') \parallel y' = f(x) \parallel y$, so $f(x') = f(x)$:
 $\Pr[\mathcal{B} \text{ wins}] = \Pr[\mathcal{A} \text{ wins}]$.

Candidate One-Way Functions

Factoring based

$$f_{\text{mult}}(x, y) = x \cdot y,$$

where x, y are **uniformly random** n -bit primes. (For arbitrary x, y , the product is even w.p. 3/4: easy to invert.)

Subset-sum based

$$f_{\text{SS}}(x_1, \dots, x_n, J) = \left(x_1, \dots, x_n, \left[\sum_{j \in J} x_j \bmod 2^n \right] \right),$$

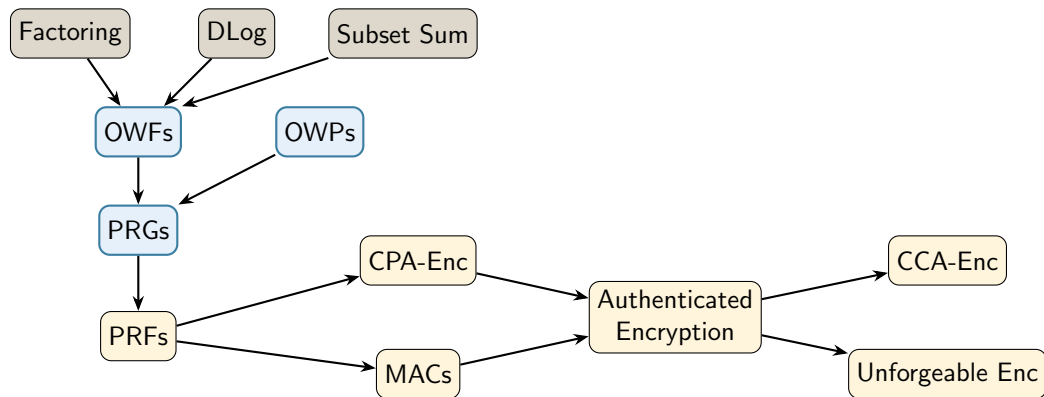
where each $x_i \in \{0, 1\}^n$ and $J \subseteq \{1, \dots, n\}$.

Discrete-log based

$$f_{p,g}(x) = [g^x \bmod p],$$

where p is a large prime, g is a **generator** of $\mathbb{Z}_p^*$, and $x \leftarrow \mathbb{Z}_{p-1}$.

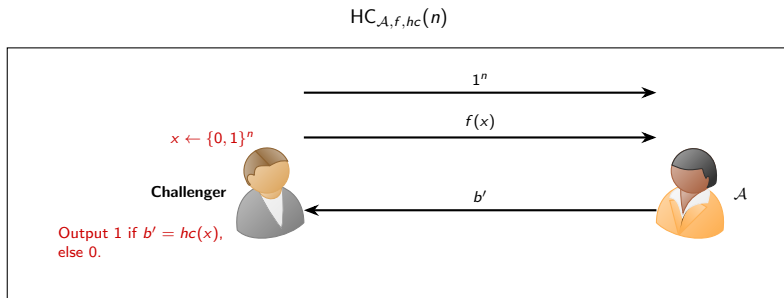
Theoretical Landscape



Where we are

Today: $\text{OWP} \Rightarrow \text{Hardness Concentration Bit} \Rightarrow \text{PRG}$ (any polynomial stretch). Next lecture: $\text{PRG} \Rightarrow \text{PRF}$.

$HC_{\mathcal{A},f,hc}$: Challenger vs. Adversary



Remark

If f is a permutation, $f(x)$ determines x and hence the bit $hc(x)$ — yet no PPT $\mathcal{A}$ guesses it noticeably better than a coin flip. The hardness of inverting f is concentrated in one bit.

Hardness Concentration Bits

Idea

Concentrate the hardness of a OWP into *one bit*.

Game $\text{HC}_{\mathcal{A},f,hc}(n)$

- 1 $x \leftarrow \{0, 1\}^n$. $\mathcal{A}$ is given 1^n and $f(x)$.
- 2 $\mathcal{A}$ outputs a bit b' .
- 3 Output 1 if $b' = hc(x)$, and 0 otherwise.

Definition 2 (Hardness Concentration Bit)

$hc : \{0, 1\}^* \rightarrow \{0, 1\}$ is a **Hardness Concentration Bit** of f if hc is computable in polynomial time and for all PPT $\mathcal{A}$: $\Pr [\text{HC}_{\mathcal{A},f,hc}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$.

Can We Take the XOR of All Input Bits?

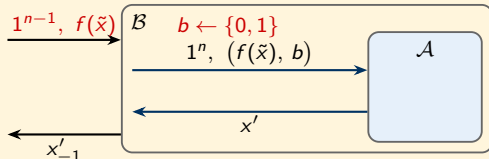
Attempt

Try the predicate $hc(x) = \bigoplus_i x_i = \sum_i x_i \bmod 2$, for every OWF.

Bad idea!

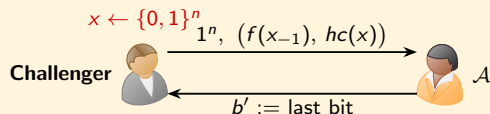
Given a OWF f , let $g(x) = (f(x_{-1}), hc(x))$, where x_{-1} is x with its last bit removed.

g is one-way: reduction to f



Why: $hc(x)$ is uniform and independent of x_{-1} (thanks to x 's last bit): $\mathcal{A}$ sees its own game. If $\mathcal{A}$ wins, $f(x'_{-1}) = f(\tilde{x})$, so $\Pr[\mathcal{B} \text{ wins}] = \Pr[\mathcal{A} \text{ wins}]$.

The attack: hc is predictable from $g(x)$



$b' = hc(x)$: $\mathcal{A}$ wins $\text{HC}_{\mathcal{A}, g, hc}$ with probability 1. So the XOR of all bits is **not** a Hardness Concentration Bit of every OWF.

Practice Problem

Problem

Construct a OWF g for which *no* single input bit is a Hardness Concentration Bit. (Assume the existence of a OWF f .)

Hint

$$g(x, i) = (f(x_{-i}), i, x_i),$$

where x_{-i} is x with its i -th bit removed.

To predict bit j of x : if the given i equals j , output the revealed x_i ; otherwise guess. This wins with probability $\frac{1}{n} \cdot 1 + (1 - \frac{1}{n}) \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{2n}$ — not $\frac{1}{2} + \text{negl}(n)$.

Constructing the Hardness Concentration Bit Function

- Given a OWP f : we don't know whether *it* has a Hardness Concentration Bit function. (Open problem.)
- Instead, given a OWP f , we build a related OWP g that *does* have a Hardness Concentration Bit function hc .

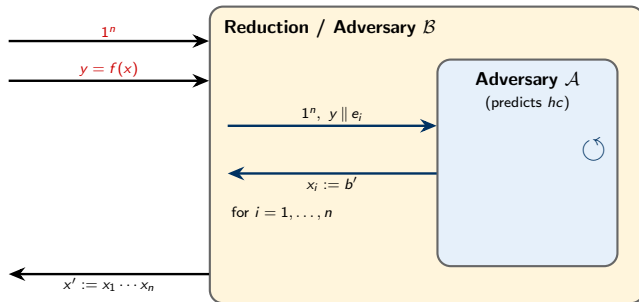
Goldreich–Levin construction

$$g(x, r) = f(x) \parallel r, \quad |x| = |r| = n,$$
$$hc(x, r) = \langle x, r \rangle = \sum_{i=1}^n x_i \cdot r_i \bmod 2.$$

Theorem 3 (Goldreich–Levin)

If f is a OWP, then hc is a Hardness Concentration Bit of g .

Warm-Up: A Perfect Predictor



Why it works

With $e_i = 0^{i-1}10^{n-i}$, $hc(x, e_i) = \langle x, e_i \rangle = x_i$. If $\mathcal{A}$ is **always** right, $\mathcal{B}$ recovers x and wins $\text{Invert}_{\mathcal{B}, f}$ with probability 1.

A Predictor That Is Only Usually Right

What's wrong if the probability is < 1 ?

$\mathcal{A}$ is right on *average* over a uniform r . It may always fail on the n special inputs e_i .

Key insight

$\mathcal{B}$ can run $\mathcal{A}$ **many more times**, on related random-looking inputs, and combine the answers to **extract** x .

- E.g. ask about r and $r \oplus e_i$ for random r : $\langle x, r \rangle \oplus \langle x, r \oplus e_i \rangle = x_i$. Repeat with fresh r 's and take the **majority**.
- Making this work when $\mathcal{A}$ is right with probability only $\frac{1}{2} + \epsilon$ (Goldreich–Levin) needs sophisticated mathematical machinery.

We will skip the proof

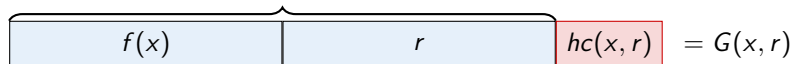
We take the Goldreich–Levin theorem as given.

Construction of a PRG (One Extra Bit) from a OWP

Given a OWP f , let $g(x, r) = f(x) \parallel r$ and $hc(x, r) = \langle x, r \rangle$ as before. Define

$$G(x, r) := f(x) \parallel r \parallel hc(x, r), \quad G : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n+1}.$$

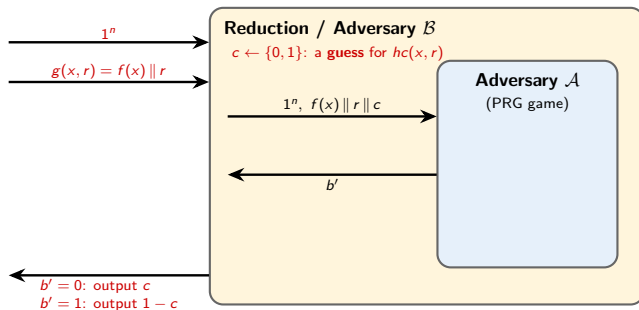
uniform in $\{0, 1\}^{2n}$, since f is a permutation and r is uniform


$$\overbrace{f(x) \parallel r} \parallel hc(x, r) = G(x, r)$$


$$\text{uniform } U_{2n} \parallel U_1 = U_{2n+1}$$

- The first $2n$ bits are distributed **exactly** as in U_{2n+1} .
- So distinguishing $G(U_{2n})$ from U_{2n+1} means telling $hc(x, r)$ from a random bit, given $g(x, r)$: **predicting** hc .

G Is a PRG: Proof by Reduction



Given / To prove

Given: $\Pr[\text{PRG}_{\mathcal{A}, G}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$ for non-negligible ϵ .

To prove: $\Pr[\text{HC}_{\mathcal{B}, g, hc}(n) = 1] \geq \frac{1}{2} + \delta(n)$ for non-negligible δ .

Intuition: if the guess is right, $f(x) \parallel r \parallel c = G(x, r)$. So $b' = 0$ ($\mathcal{A}$ says “pseudorandom”) is evidence that $c = hc(x, r)$; $b' = 1$ (“uniform”) suggests flipping it.

G Is a PRG: Two Cases

Let $p_0 = \Pr[b' = 0 \mid \mathcal{A} \text{ gets } G(x, r)]$ and $p_1 = \Pr[b' = 1 \mid \mathcal{A} \text{ gets } U_{2n+1}]$, so $\frac{1}{2}(p_0 + p_1) \geq \frac{1}{2} + \epsilon$.

Case $c = hc(x, r)$

$\mathcal{A}$ gets exactly $G(x, r)$. $\mathcal{B}$ is right iff $b' = 0$: probability p_0 .

Case $c \neq hc(x, r)$

$\mathcal{A}$ gets $f(x) \| r \| \overline{hc(x, r)}$. $\mathcal{B}$ is right iff $b' = 1$: probability q , say.

What is q ?

U_{2n+1} is $f(x) \| r \| c$ with a uniform c : half the time it is $G(x, r)$, half the time the flipped string. So

$$p_1 = \frac{1}{2}(1 - p_0) + \frac{1}{2}q, \quad \text{i.e.} \quad q = 2p_1 + p_0 - 1.$$

G Is a PRG: Putting It Together

Success probability of $\mathcal{B}$

$$\begin{aligned}\Pr [\text{HC}_{\mathcal{B},g,hc}(n) = 1] &= \frac{1}{2} p_0 + \frac{1}{2} q && (c \text{ is uniform}) \\ &= \frac{1}{2} p_0 + \frac{1}{2} (2p_1 + p_0 - 1) \\ &= p_0 + p_1 - \frac{1}{2} \\ &\geq 2\left(\frac{1}{2} + \epsilon(n)\right) - \frac{1}{2} = \frac{1}{2} + 2\epsilon(n),\end{aligned}$$

contradicting that hc is a Hardness Concentration Bit for g (Goldreich–Levin). $\square$

Remark (running time)

$\mathcal{B}$ runs $\mathcal{A}$ once and flips one coin: PPT.

From One Extra Bit to Two

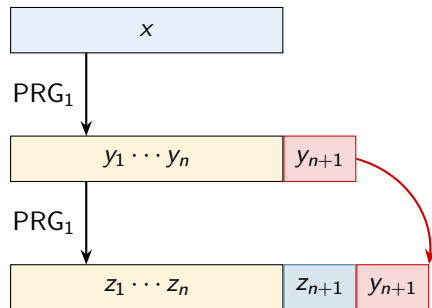
Let $\text{PRG}_1 : \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ be a PRG with one bit of stretch. Build $\text{PRG}_2 : \{0, 1\}^n \rightarrow \{0, 1\}^{n+2}$:

$\text{PRG}_2(x_1 \cdots x_n)$

- 1 $y_1 \cdots y_n \textcolor{red}{y}_{n+1} := \text{PRG}_1(x_1 \cdots x_n)$.
- 2 $z_1 \cdots z_n \textcolor{blue}{z}_{n+1} := \text{PRG}_1(y_1 \cdots y_n)$.
- 3 Output $z_1 \cdots z_n \textcolor{blue}{z}_{n+1} \textcolor{red}{y}_{n+1}$.

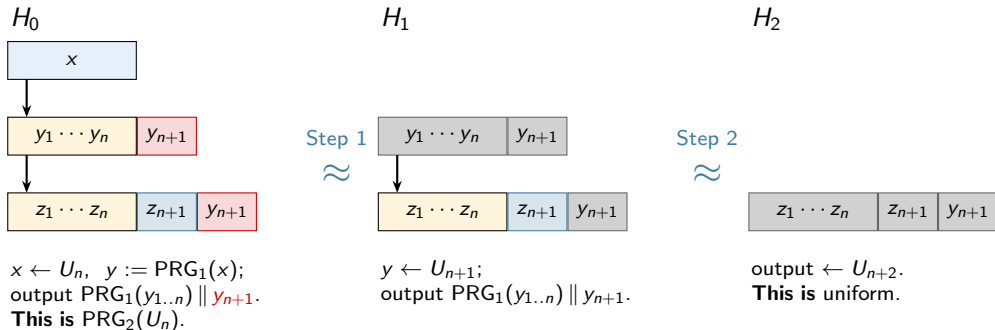
Idea

Set **one bit** aside; the other n bits look like a fresh seed, so stretch them **again**.



PRG₂ Is a PRG: Three Hybrids

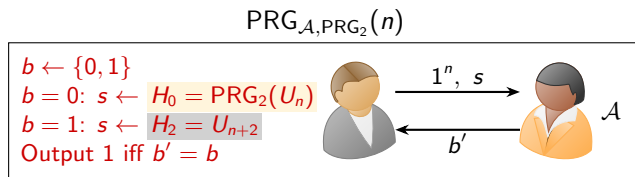
Replace the two PRG₁ calls by **truly uniform** strings (grey), **one call at a time**:



The PRG game for PRG₂ asks $\mathcal{A}$ to tell H_0 from H_2 . Each step changes **one** PRG₁ output into a uniform string.

PRG₂ Is a PRG: Suppose Not

Suppose some PPT $\mathcal{A}$ has $\Pr[\text{PRG}_{\mathcal{A}, \text{PRG}_2}(n) = 1] \geq \frac{1}{2} + \epsilon(n)$ for a non-negligible ϵ .



Write $\Pr[\mathcal{A}(H) = 1]$ for the probability that $\mathcal{A}$ outputs 1 on 1^n and $s \leftarrow H$. Since b is uniform:

$$\begin{aligned}\Pr[\text{PRG}_{\mathcal{A}, \text{PRG}_2}(n) = 1] &= \frac{1}{2} \Pr[\mathcal{A}(H_0) = 0] + \frac{1}{2} \Pr[\mathcal{A}(H_2) = 1] \\ &= \frac{1}{2} (1 - \Pr[\mathcal{A}(H_0) = 1]) + \frac{1}{2} \Pr[\mathcal{A}(H_2) = 1] \\ &= \frac{1}{2} + \frac{1}{2} (\Pr[\mathcal{A}(H_2) = 1] - \Pr[\mathcal{A}(H_0) = 1]).\end{aligned}$$

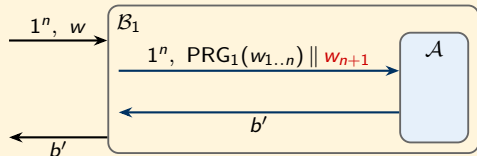
So $\mathcal{A}$ says 1 non-negligibly more often on H_2 than on H_0

$$\Pr[\mathcal{A}(H_2) = 1] - \Pr[\mathcal{A}(H_0) = 1] \geq 2\epsilon(n).$$

Each Step Is a Reduction to PRG_1

Each $\mathcal{B}_j$ plays the PRG game for PRG_1 , builds a hybrid from its challenge w , gives it to $\mathcal{A}$, and outputs $\mathcal{A}$'s guess b' .

$\mathcal{B}_1$ (Step 1): its challenge w plays y

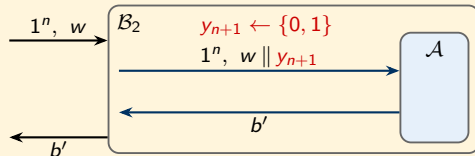


$b = 0, w \leftarrow \text{PRG}_1(U_n)$: $\mathcal{A}$ gets H_0 .

$b = 1, w \leftarrow U_{n+1}$: $\mathcal{A}$ gets H_1 . Same calculation as for $\mathcal{A}$:

$$\Pr[\text{PRG}_{\mathcal{B}_1, \text{PRG}_1}(n) = 1] = \frac{1}{2} + \frac{1}{2} (\Pr[\mathcal{A}(H_1) = 1] - \Pr[\mathcal{A}(H_0) = 1]).$$

$\mathcal{B}_2$ (Step 2): its challenge w plays z



$b = 0, w \leftarrow \text{PRG}_1(U_n)$: $\mathcal{A}$ gets H_1 .

$b = 1, w \leftarrow U_{n+1}$: $\mathcal{A}$ gets H_2 . Same calculation as for $\mathcal{A}$:

$$\Pr[\text{PRG}_{\mathcal{B}_2, \text{PRG}_1}(n) = 1] = \frac{1}{2} + \frac{1}{2} (\Pr[\mathcal{A}(H_2) = 1] - \Pr[\mathcal{A}(H_1) = 1]).$$

PRG₂: Putting It Together

From $\mathcal{A}$: $\Pr[\mathcal{A}(H_2) = 1] - \Pr[\mathcal{A}(H_0) = 1] \geq 2\epsilon(n)$. Add and subtract $\Pr[\mathcal{A}(H_1) = 1]$:

$$\begin{aligned} 2\epsilon(n) &\leq \Pr[\mathcal{A}(H_2) = 1] - \Pr[\mathcal{A}(H_0) = 1] \\ &= \underbrace{(\Pr[\mathcal{A}(H_1) = 1] - \Pr[\mathcal{A}(H_0) = 1])}_{\text{Step 1}} + \underbrace{(\Pr[\mathcal{A}(H_2) = 1] - \Pr[\mathcal{A}(H_1) = 1])}_{\text{Step 2}}. \end{aligned}$$

Two numbers that add up to at least $2\epsilon(n)$: **at least one** of them is $\geq \epsilon(n)$ (even if the other is negative). Say Step j . By the previous slide,

$$\Pr[\text{PRG}_{\mathcal{B}_j, \text{PRG}_1}(n) = 1] = \frac{1}{2} + \frac{1}{2} \cdot (\text{Step } j) \geq \frac{1}{2} + \frac{\epsilon(n)}{2},$$

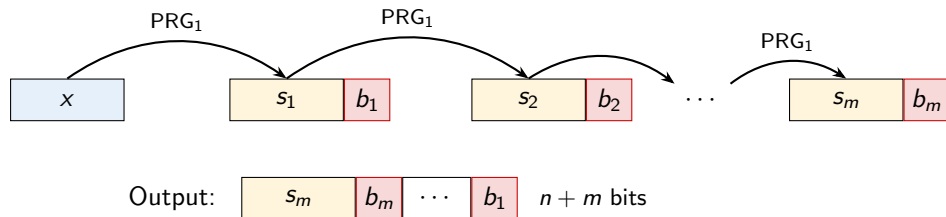
contradicting that PRG_1 is a PRG. □

Remark (running time)

Each $\mathcal{B}_j$ runs $\mathcal{A}$ once, plus one PRG_1 call or one coin flip: PPT.

Getting More Bits: PRG_m

Keep going: each round sets aside one bit and re-stretches the remaining n .



With $m = n$: a **length-doubling** PRG $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ — the starting point for PRFs, next lecture.

Summary

- **One-way functions:** easy to compute, hard to invert (game $\text{Invert}_{\mathcal{A},f}(n)$) — the foundation: $\text{OWF} \Rightarrow \text{PRG} \Rightarrow \text{PRF} \Rightarrow \dots$; the hardness can hide anywhere in the output. Candidates: factoring, subset sum, discrete log.
- **Hardness Concentration Bit:** a predicate hc that no PPT $\mathcal{A}$ predicts from $f(x)$ (game $\text{HC}_{\mathcal{A},f,hc}(n)$). The XOR of all bits fails; the **inner product** $hc(x, r) = \langle x, r \rangle$ works for $g(x, r) = f(x) \| r$ (Goldreich–Levin).
- **PRG from OWP:** $G(x, r) = f(x) \| r \| hc(x, r)$ stretches by one bit. *Proved:* a distinguisher for G becomes a predictor for hc .
- **Stretch amplification:** set one bit aside and re-stretch the rest; iterating gives any polynomial stretch. *Proved* for PRG_2 : two hybrid steps, each a reduction to PRG_1 .
- **Next lecture:** PRFs from a length-doubling PRG (GGM).